Part 1
Acme Games. A small video game development studio that insists upon self-hosting official dedicated servers for their products. 65 employees, approx. breakdown: 50 developers, 8 artists, 7 admin/IT. Especially critical assets include proprietary source code, pre-release builds, art assets, and internal tools. Because the company hosts its own public-facing game servers, those servers become an additional attack surface.

Major cybersecurity concerns: theft or leakage of source code and pre-release builds, compromise of developer workstations, unauthorized access to internal systems, and abuse or pivot from the public game servers into the internal network.

Part 2
- Developers need high-performance authenticated access to source control, build systems, and shared asset repositories.
- Official dedicated game servers must be reachable by external players while remaining strictly isolated from the internal network.
- Selected users (build engineers and administrators) need secure methods for remotely managing servers and infrastructure.
- All security-relevant activity (authentication events, administrative actions, and access to source code/assets) must be logged and retained.
- Employees need secure Internet access for research, tools, and collaboration without exposing internal systems.
- Public-facing services must be separated from internal systems by clear security boundaries.
- Sensitive resources (source control, builds, administrative interfaces) must not be accessible to every user or to untrusted networks.

Part 3

Asset	**Purpose**	**Importance**	**Security Concern**
Dev workstations	High end coding, compiling, testing	High	Endpoint compromise
Windows Server	Authentication, file shares, internal services	High	Privilege escalation
Linux game server(s)	Official dedicated multiplayer servers	High	Compromise leading to service abuse
Source control/asset repository	Store proprietary code and builds	Critical	Unauthorized access, data exfiltration
Firewall	Boundary control and segmentation	High	Misconfiguration or bypass

Management	Secure admin access	High	Lateral movement
Logging/monitoring system	Centralized visibility	High	Blind spots

Part 4

- Windows workstations: Required for the developer and artist workflow (IDEs, compilers, art tools, local testing).
- Windows Server: Provides Active Directory for centralized authentication, file services for source/assets, and other internal services.
- Linux servers: Host the official dedicated multiplayer game servers that players connect to.
- Active Directory: Central authentication and authorization for users and systems.
- DNS & DHCP: Name resolution and automated addressing for the User network (servers will primarily use static addresses).
- File services: Shared repositories for source code, builds, and art assets.
- Logging & monitoring: Windows Event Forwarding + Linux syslog/journald collection for security visibility.
- Firewall: Enforces segmentation and controls traffic between zones and the Internet.

Part 5

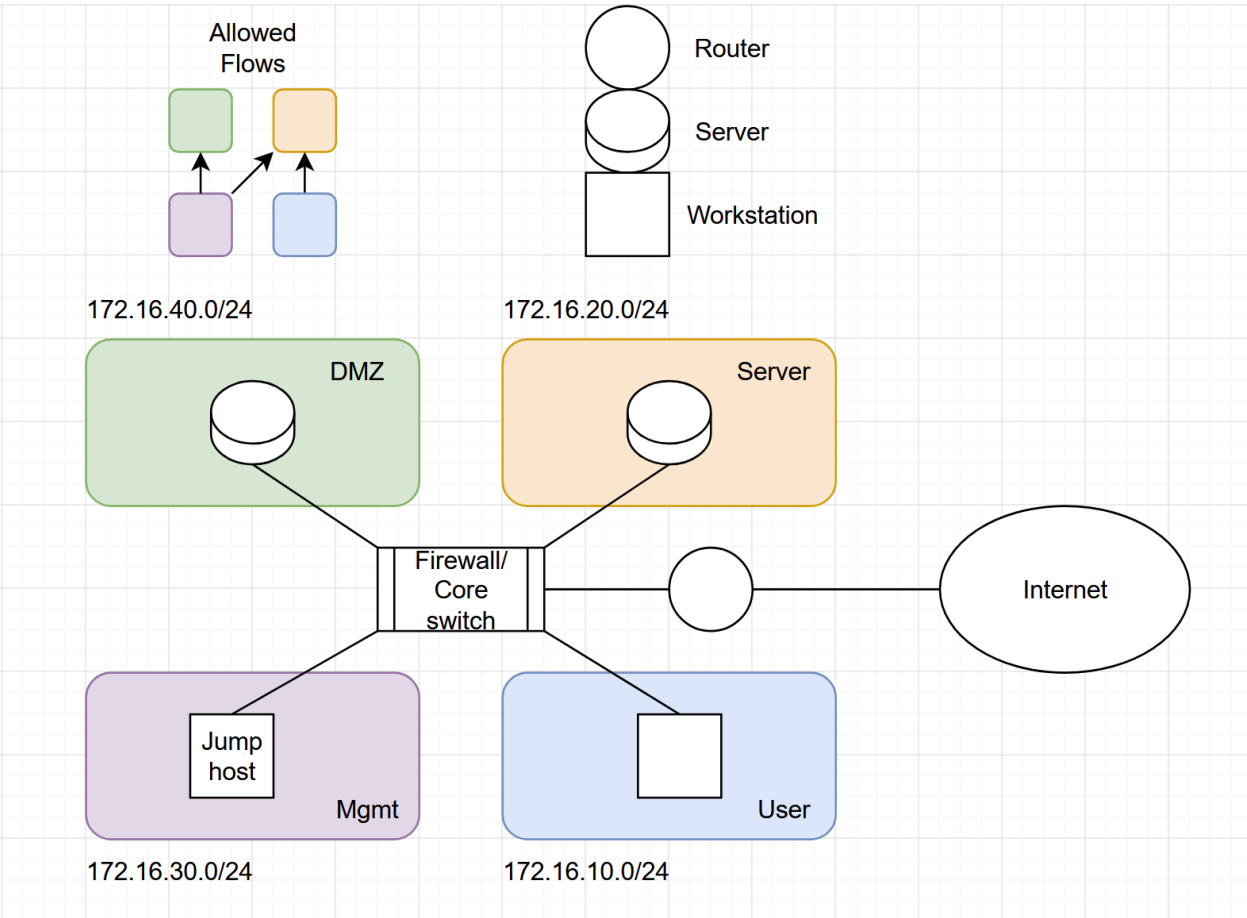
Name	**Purpose**	**Who lives there**	**Who can access**	**Key restrictions**
User	Designated for end users, employees	Employee Workstations	Employees	Selective paths to Server only
Server	Designated for servers	Windows Servers	User, Mgmt	Selective paths to Mgmt and User only
Mgmt	Used by admin to configure monitor and control the network	Admin workstations	Authorized administrators	No direct access from User > Mgmt. Mgmt reaches Server and DMZ via controlled paths
DMZ	Public facing network, locked down	Linux game server	Mgmt	Only accept player traffic on specific ports

Part 6

Name	IPv4	Gateway	Purpose
User	172.16.10.0/24	172.16.10.1	Employees
Server	172.16.20.0/24	172.16.20.1	Internal services
Mgmt	172.16.30.0/24	172.16.30.1	Admin access
DMZ	172.16.40.0/24	172.16.40.1	Locked down public facing server

These private ranges are non-overlapping, easy to remember, and leave room for modest growth within each /24. Servers will use static addressing, user networks will use DHCP.

Part 7



Part 8

- Administrative access must be separated from standard user activity (jump hosts/mgmt)
- Unnecessary network services must be disabled on all systems.
- Communication between security zones must be restricted to business need only (explicit

firewall rules).
- Sensitive services and data (source control, builds) must not be directly exposed to untrusted networks.

- Systems must use appropriate authentication controls (AD + MFA where feasible, strong local policies).
- Security-relevant events must be logged and protected from tampering.
- Systems must be updated and hardened before formal security testing.
- Public-facing game servers must run with least privilege and minimal attack surface.

Part 9

- Credential compromise > workstations, AD > lateral movement, source access > strong auth, least privilege, monitoring
- Vulnerable services on game servers > DMZ > player abuse or pivot > hardening, minimal services, network restrictions
- Data exposure > source repositories > competitive damage > access controls, encryption at rest/transit
- Insider misuse > any high value system > leakage of builds > logging and separation of duties
- Misconfiguration > firewall or servers > bypass of controls > change control

Part 10

- Windows Event Logs (Security, System, Application)
- Linux syslog / journald from game servers
- Firewall logs
- Authentication logs
- Vulnerability scan results
- Configuration baselines / hardening checklists
- Command output and screenshots as evidence
- Network captures (Wireshark) for selected flows if needed

If suspicious activity occurred, I would look first at authentication logs and firewall deny logs because failed or unusual authentication attempts combined with inter-zone traffic would be the earliest reliable indicators of credential abuse or attempted lateral movement.

Part 11

Week 3 - Architecture and initial setup
Week 4 - Build and configure
Week 5 - Secure and harden
Week 6 - Testing
Week 7 - Further testing, monitor and improve
Week 8 - Finalize

Part 12

These virtual machines have been created in VMWare Workstation Pro:
- Windows server (Server)
- Linux server (DMZ)
- Windows 11 workstation (user)
- Windows 11 jump host (mgmt)